

AMAZING

SCIENCE FICTION STORIES

TRANSIENT

A Novel by
Ward Moore

FEBRUARY

35¢



MAC 2

AMAZING

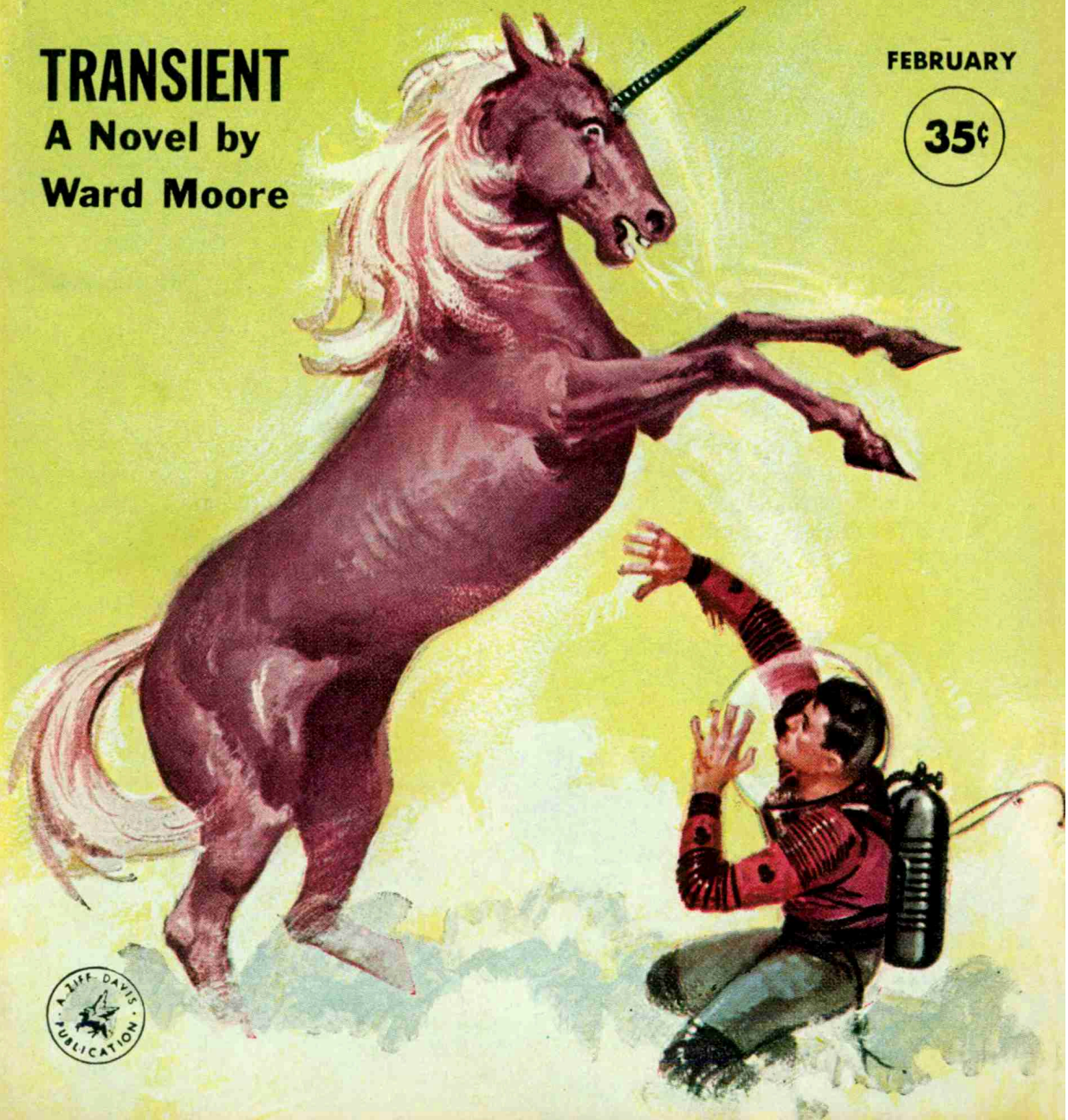
SCIENCE FICTION STORIES

TRANSIENT

A Novel by
Ward Moore

FEBRUARY

35¢



The Project Gutenberg eBook of A long way back

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: A long way back

Author: Ben Bova

Illustrator: Leo Summers

Release date: December 30, 2023 [eBook #72553]

Language: English

Original publication: New York, NY: Ziff-Davis Publishing Company, 1960

Other information and formats: www.gutenberg.org/ebooks/72553

Credits: Greg Weeks, Mary Meehan and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK A LONG WAY
BACK ***

A LONG WAY BACK

By BEN BOVA

ILLUSTRATED by SUMMERS

*He held the future of the world
in his numbed hands. And from 22,500
miles out, he made the gamble.*

**[Transcriber's Note: This etext was produced from
Amazing Stories February 1960.
Extensive research did not uncover any evidence that
the U.S. copyright on this publication was renewed.]**

Tom woke slowly, his mind groping back through the hypnosis. He found himself looking toward the observation port, staring at stars and blackness.

The first man in space, he thought bitterly.

He unstrapped himself from the acceleration seat, feeling a little wobbly in free fall.

The hypnotic trance idea worked, all right.

The last thing Tom remembered was Arnoldsson putting him under, here in the rocket's compartment, the old man's sad soft eyes and quiet voice. Now 22,500 miles out, Tom was alone except for what Arnoldsson had planted in his mind for post-hypnotic suggestion to recall. The hypnosis had helped him pull through the blastoff unhurt and even protected him against the vertigo of weightlessness.

Yeah, it's a wonderful world, Tom muttered acidly.

He got up from the seat cautiously, testing his coordination against zero gravity. His magnetic boots held to the deck satisfactorily.

He was lean and wiry, in his early forties, with a sharp angular face and dark, somber eyes. His hair had gone dead white years ago. He was encased up to his neck in a semi-flexible space suit; they had squirmed him into it Earth-side because there was no room in the cramped cabin to put it on.

Tom glanced at the tiers of instrument consoles surrounding his seat—no blinking red lights, everything operating normally. *As if I could do anything about it if they went wrong.* Then he leaned toward the observation port, straining for a glimpse of the satellite.

The satellite.

Five sealed packages floating within a three-hundred foot radius of emptiness, circling the Earth like a cluster of moonlets. Five pieces sent up in five robot rockets and placed in the same orbit, to wait for a human intelligence to assemble them into a power-beaming satellite.

Five pieces orbiting Earth for almost eighteen years; waiting for nearly eighteen years while down below men blasted themselves and their cities

and their machines into atoms and forgot the satellite endlessly circling, waiting for its creators to breathe life into it.

The hope of the world, Tom thought. And little Tommy Morris is supposed to make it work ... and then fly home again. He pushed himself back into the seat. *Jason picked the wrong man.*

"Tom! Tom, can you hear me?"

He turned away from the port and flicked a switch on the radio console.

"Hello Ruth. I can hear you."

A hubbub of excitement crackled through the radio receiver, then the girl's voice: "Are you all right? Is everything...."

"Everything's fine," Tom said flatly. He could picture the scene back at the station—dozens of people clustered around the jury-rigged radio, Ruth working the controls, trying hard to stay calm when it was impossible to, brushing back that permanently displaced wisp of brown hair that stubbornly fell over her forehead.

"Jason will be here in a minute," she said. "He's in the tracking shack, helping to calculate your orbit."

Of course Jason will be here, Tom thought. Aloud he said, "He needn't bother. I can see the satellite packages; they're only a couple of hundred yards from the ship."

Even through the radio he could sense the stir that went through them.

Don't get your hopes up, he warned silently. *Remember, I'm no engineer. Engineers are too valuable to risk on this job. I'm just a tool, a mindless screwdriver sent here to assemble this glorified tinkertoy. I'm the muscle, Arnoldsson is the nerve link, and Jason is the brain.*

Abruptly, Jason's voice surged through the radio speaker, "We did it, Tom! We did it!"

No, Tom thought, you did it, Jason. This is all your show.

"You should be able to see the satellite components," Jason said. His voice was excited yet controlled, and his comment had a ring of command in it.

"I've already looked," Tom answered. "I can see them."

"Are they damaged?"

"Not as far as I can see. Of course, from this distance...."

"Yes, of course," Jason said. "You'd better get right outside and start working on them. You've only got forty-eight hours worth of oxygen."

"Don't worry about me," Tom said into the radio. "Just remember your end of the bargain."

"You'd better forget that until you get back here."

"I'm not forgetting anything."

"I mean you must concentrate on what you're doing up there if you expect to get back alive."

"When I get back we're going to explore the bombed-out cities. You promised that. It's the only reason I agreed to this."

Jason's voice stiffened. "My memory is quite as good as yours. We'll discuss the expedition after you return. Now you're using up valuable time ... and oxygen."

"Okay. I'm going outside."

Ruth's voice came back on: "Tom, remember to keep the ship's radio open, or else your suit radio won't be able to reach us. And we're all here ... Dr. Arnoldsson, Jason, the engineers ... if anything comes up, we'll be right here to help you."

Tom grinned mirthlessly. *Right here: 22,500 miles away.*

"Tom?"

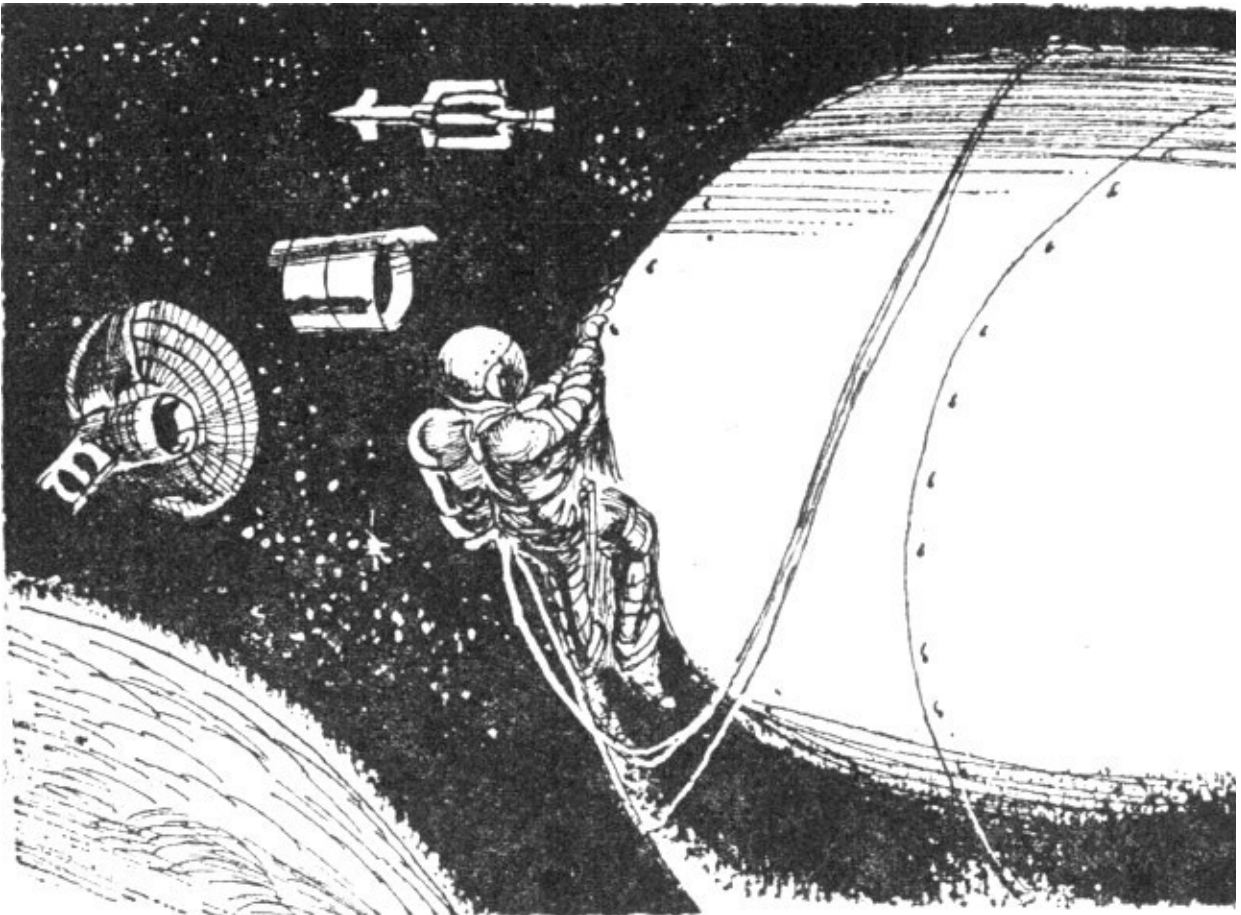
"Yes Ruth."

"Good luck," she said. "From all of us."

Even Jason? he wanted to ask, but instead said merely, "Thanks."

He fitted the cumbersome helmet over his head and sealed it to the joints on his suit. A touch of a button on the control panel pumped the compartment's air into storage cylinders. Then Tom stood up and unlocked the hatch directly over his seat.

Reaching for the handholds just outside the hatch, he pulled himself through, and after a weightless comic ballet managed to plant his magnetized boots on the skin of the ship. Then, standing, he looked out at the universe.



Oddly, he felt none of the overpowering emotion he had once expected of this moment. Grandeur, terror, awe—no, he was strangely calm. The stars were only points of light on a dead-black background; the Earth was a fat crescent patched with colors; the sun, through his heavily-tinted visor, was like the pictures he had seen at planetarium shows, years ago.

As he secured a lifeline to the grip beside the hatch, Tom thought that he felt as though someone had stuck a reverse hypodermic into him and drained away all his emotions.

Only then did he realize what had happened. Jason, the engineer, the leader, the man who thought of everything, had made Arnoldsson condition his mind for this. No gaping at the universe for the first man in space, too much of a chance to take! There's a job to be done and no time for human frailty or sentiment.

Not even that, Tom said to himself. *He wouldn't even allow me one moment of human emotion.*

But as he pushed away from the ship and floated ghost-like toward the largest of the satellite packages, Tom twisted around for another look at Earth.

I wonder if she looked that way before the war?

Slowly, painfully, men had attempted to rebuild their civilization after the war had exhausted itself. But of all the things destroyed by the bombs and plagues, the most agonizing loss was man's sources of energy.

The coal mines, the oil refineries, the electricity-generating plants, the nuclear power piles ... all shattered into radioactive rubble. There could be no return to any kind of organized society while men had to scavenge for wood to warm themselves and to run their primitive machines.

Then someone had remembered the satellite.

It had been designed, before the war, to collect solar energy and beam it to a receiving station on Earth. The satellite packages had been fired into a 24-hour orbit, circling the Earth over a fixed point on the Equator. The receiving station, built on the southeastern coast of the United States, saw the five units as a single second-magnitude star, low on the horizon all year, every year.

Of course the packages wavered slightly in their orbits, but not enough in eighteen years to spread very far apart. A man could still put them together into a power-beaming satellite.

If he could get there.

And if they were not damaged.

And if he knew how to put them together.

Through months that stretched into years, over miles of radioactive wilderness, on horseback, on carts, on foot, those who knew about the satellite spread the word, carefully, secretly, to what was left of North America's scientists and engineers. Gradually they trickled into the once-abandoned settlement.

They elected a leader: Jason, the engineer, one of the few men who knew anything about rockets to survive the war and the lunatic bands that hunted down anyone suspected of being connected with pre-war science.

Jason's first act was to post guards around the settlement. Then he organized the work of rebuilding the power-receiving station and a man-carrying rocket.

They pieced together parts of a rocket and equipment that had been damaged by the war. What they did not know, they learned. What they did not have, they built or cannibalized from ruined equipment.

Jason sent armed foragers out for gasoline, charcoal and wood. They built a ramshackle electricity generator. They planted crops and hunted the small game in the local underbrush. A major celebration occurred whenever a forager came back towing a stray cow or horse or goat.

They erected fences around the settlement, because more than once they had to fight off the small armies of looters and anti-scientists that still roved the countryside.

But finally they completed the rocket ... after exhausting almost every scrap of material and every ounce of willpower.

Then they picked a pilot: Thomas H. Morris, age 41, former historian and teacher. He had arrived a year before the completion of the rocket after walking 1,300 miles to find the settlement; his purpose was to organize some of the scientists and explore the bombed-out cities to see what could be salvaged out of man's shattered heritage.

But Tom was ideal for the satellite job: the right size—five-six and one-hundred thirty pounds; no dependents—wife and two sons dead of radiation sickness. True, he had no technical background whatsoever; but with Arnoldsson's hypnotic conditioning he could be taught all that it was necessary for him to know ... maybe.

Best of all, though, he was thoroughly expendable.

So Jason made a deal with him. There could be no expeditions into the cities until the satellite was finished, because every man was needed at the settlement. And the satellite could not be finished until someone volunteered to go up in the rocket and assemble it.

It was like holding a candy bar in front of a small child. He accepted Jason's terms.

The Earth turned, and with it the tiny spark of life alone in the emptiness around the satellite. Tom worked unmindful of time, his eyes and hands following Jason's engineering commands through Arnoldsson's post-hypnotic directions, with occasional radio conferences.

But his conscious mind sought refuge from the strangeness of space, and he talked almost constantly into his radio while he worked, talked about anything, everything, to the girl on the other end of the invisible link.

"... and once the settlement is getting the power beamed from this contraption, we're going to explore the cities. Guess we won't be able to get very far inland, but we can still tackle Washington, Philadelphia and New York ... plenty for us there."

Ruth asked, "What were they like before the war?"

"The cities? That's right, you're too young to remember. They were big, Ruth, with buildings so tall people called them skyscrapers." He pulled a wrench from its magnetic holder in the satellite's self-contained tool bin. "And filled with life. Millions of people lived in each one ... all the people we have at the settlement could have lived on one floor of a good-sized hotel...."

"What's a hotel?"

Tom grinned as he tugged at a pipe fitting. "You'll find out when you come with us ... you'll see things you could never imagine."

"I don't know if I'll come with you."

He looked up from his work and stared Earthward. "Why?"

"Well ... Jason ... he says there isn't much left to see. And it's all radioactive and diseased."

"Nonsense."

"But Jason says...."

Tom snorted. "Jason hasn't been out of the settlement for six years. I walked from Chicago to the settlement a year ago. I went through a dozen cities ... they're wrecked, and the radioactivity count was higher than it is here at the settlement, but it's not high enough to be dangerous."

"And you want to explore those cities; why?"

"Let's just say I'm a historian," Tom answered while his hands manipulated complex wiring unconsciously, as though they belonged not to him but to some unseen puppeteer.

"I don't understand," Ruth said.

"Look—those cities hold mankind's memory. I want to gather up the fragments of civilization before the last book is used for kindling and the last machine turns to rust. We need the knowledge in the cities if we expect to rebuild a civilization...."

"But Jason and Dr. Arnoldsson and the engineers—they know all about...."

"Jason and the engineers," Tom snapped. "They had to stretch themselves to the breaking point to put together this rocket from parts that were already manufactured, waiting for them. Do you think they'd know how to build a city? Dr. Arnoldsson is a psychiatrist; his efforts at surgery are pathetic. Have you ever seen him try to set a broken leg? And what about agriculture? What about tool-making or mining or digging wells, even ... what about education? How many kids your own age can read or write?"

"But the satellite...."

"The satellite won't be of any use to people who can't work the machines. The satellite is no substitute for knowledge. Unless something is done, your grandchildren will be worshipping the machines, but they won't know how to repair them."

"No...."

"Yes, Ruth," he insisted.

"No," she whispered, her voice barely audible over the static-streaked hum in his earphones. "You're wrong, Tom. You're wrong. The satellite will send us the power we need. Then we'll build our machines and teach our children."

How can you teach what you don't know? Tom wanted to ask, but didn't. He worked without talking, hauling the weightless tons of satellite packages into position, electronically welding them together, splicing wiring systems too intricate for his conscious mind to understand.

Twice he pulled himself back along the lifeline into the ship for capsule meals and stimulants.

Finally he found himself staring at his gloved hands moving industriously within the bowels of one of the satellite packages. He stopped, suddenly aware that it was piercingly cold and totally dark except for the lamp on his helmet.

He pushed away from the unfinished satellite. Two of the packages were assembled now. The big parabolic mirror and two other uncrated units hung nearby, waiting impassively.

Tom groped his way back into the ship. After taking off his helmet and swallowing a couple of energy pills he said to the ship's radio:

"What time is it?" The abrupt sound of his own voice half-startled him.

"Nearly four a.m." It was Jason.

"Earth's blotted out the sun," Tom muttered. "Getting damned cold in here."

"You're in the ship?"

"Yes. It got too cold for the suit."

"Turn up the ship's heaters," Jason said. "What's the temperature in there?"

Tom glanced at the thermometer as he twisted the thermostat dial as far as it would go. "Forty-nine," he answered.

He could sense Jason nod. "The heaters are on minimum power automatically unless you turn them up. It'll warm you up in a few seconds. How's the satellite?"

Tom told him what remained to be done.

"You're not even half through yet." Jason's voice grew fainter and Tom knew that he was doing some mental arithmetic as he thought out loud. "You've been up about twenty hours; at the rate you're going you'll need another twenty-four to finish the job. That will bring you very close to your oxygen limit."

Tom sat impassively and stared at the gray metal and colored knobs of the radio.

"Is everything going all right?" Jason asked.

"How should I know? Ask Arnoldsson."

"He's asleep. They all are."

"Except you."

"That's right," Jason said, "except me."

"How long did Ruth stay on the radio?"

"About sixteen hours. I ordered her to sleep a few hours ago."

"You're pretty good at giving orders," Tom said.

"Someone has to."

"Yeah." Tom ran a hand across his mouth. *Boy, could I use a cigarette. Funny, I haven't even thought about them in years.*

"Look," he said to the radio, "we might as well settle something right now. How many men are you going to let me have?"

"Don't you think you'd better save that for now and get back to work?"

"It's too damned cold out there. My fingers are still numb. You could have done a better job on insulating this suit."

"There are a lot of things we could have done," Jason said, "if we had the material."

"How about the expedition? How many men can I have?"

"As many as you can get," the radio voice answered. "I promised I won't stand in your way once the satellite is finished and operating."

"Won't stand in my way," Tom repeated. "That means you won't encourage anyone, either."

Jason's voice rose a trifle. "I can't encourage my people to go out and risk their lives just because you want to poke around some radioactive slag heaps!"

"You promised that if I put the satellite together and got back alive, I could investigate the cities. That was our deal."

"That's right. You can. And anyone foolish enough to accompany you can follow along."

"Jason, you know I need at least twenty-five armed men to venture out of the settlement...."

"Then you admit it's dangerous!" the radio voice crackled.

"Sure, if we meet a robber band. You've sent out enough foraging groups to know that. And we'll be travelling hundreds of miles. But it's not dangerous for the reasons you've been circulating ... radioactivity and disease germs and that nonsense. There's no danger that one of your own foraging groups couldn't handle. I came through the cities last year alone, and I made it."

Tom waited for a reply from the radio, but only the hissing and crackling of electrical disturbances answered him.

"Jason, those cities hold what's left of a world-wide civilization. We can't begin to rebuild unless we reopen that knowledge. We need it, we need it desperately!"

"It's either destroyed or radioactive, and to think anything else is self-delusion. Besides, we have enough intelligence right here at the settlement to build a new civilization, better than the old one, once the satellite is ready."

"But you don't!" Tom shouted. "You poor damned fool, you don't even realize how much you don't know."

"This is a waste of time," Jason snapped. "Get outside and finish your work."

"I'm still cold, dammit," Tom said. He glanced at the thermometer on the control console. "Jason! *It's below freezing in here!*"

"What?"

"The heating unit isn't working at all!"

"Impossible. You must have turned it off instead of on."

"I can read, dammit! It's turned as high as it'll go...."

"What's the internal thermometer reading?"

Tom looked. "Barely thirty ... and it's still going down."

"Hold on, I'll wake Arnoldsson and the electrical engineers."

Silence. Tom stared at the inanimate radio which gave off only the whines and scratches of lightning and sun and stars, all far distant from him. For all his senses could tell him, he was the last living thing in the universe.

Sure, call a conference, Tom thought. How much more work is there to be done? About twenty-four hours, he said. Another day. And another full night. Another night, this time with no heat. And maybe no oxygen, either. The heaters must have been working tonight until I pushed them up to full power. Something must have blown out. Maybe it's just a broken wire. I could fix that if they tell me how. But if it's not ... no heat tomorrow night, no heat at all.

Then Arnoldsson's voice floated up through the radio speaker: soft, friendly, calm, soothing....

The next thing Tom knew he was putting on his helmet. Sunlight was lancing through the tinted observation port and the ship was noticeably warmer.

"What happened?" he mumbled through the dissolving haze of hypnosis.

"It's all right, Tom." Ruth's voice. "Dr. Arnoldsson put you under and had you check the ship's wiring. Now he and Jason and the engineers are figuring out what to do. They said it's nothing to worry about ... they'll have everything figured out in a couple of hours."

"And I'm to work on the satellite until they're ready?"

"Yes."

"Don't call us, we'll call you."

"What?"

"Nothing."

"It's all right, Tom. Don't worry."

"Sure Ruth, I'm not worried." *That makes us both liars.*

He worked mechanically, handling the unfamiliar machinery with the engineers' knowledge through Arnoldsson's hypnotic communication.

Just like the pictures they used to show of nuclear engineers handling radioactive materials with remotely-controlled mechanical hands from behind a concrete wall. I'm only a pair of hands, a couple of opposed

thumbs, a fortunate mutation of a self-conscious simian ... but, God, why don't they call? She said it wasn't anything big. Just the wiring, probably. Then why don't they call?

He tried to work without thinking about anything, but he couldn't force his mind into stillness.

Even if I can fix the heaters, even if I don't freeze to death, I might run out of oxygen. And how am I going to land the ship? The takeoff was automatic, but even Jason and Arnoldsson can't make a pilot out of me....

"Tom?" Jason's voice.

"Yes!" He jerked to attention and floated free of the satellite.

"We've ... eh, checked what you told us about the ship's electrical system while Arnoldsson had you under the hypnotic trance...."

"And?"

"Well ... it, eh, looks as though one of the batteries gave out. The batteries feed all the ship's lights, heat, and electrical power ... with one of them out, you don't have enough power to run the heaters."

"There's no way to fix it?"

"Not unless you cut out something else. And you need everything else ... the radio, the controls, the oxygen pumps...."

"What about the lights? I don't need them, I've got the lamp on my suit helmet."

"They don't take as much power as the heaters do. It wouldn't help at all."

Tom twisted weightlessly and stared back at Earth. "Well just what the hell am I supposed to *do*?"

"Don't get excited," Jason's voice grated in his earphones. "We've calculated it all out. According to our figures, your suit will store enough heat during the day to last the night...."

"I nearly froze to death last night and the ship was heated most of the time!"

"It will get cold," Jason's voice answered calmly, "but you should be able to make it. Your own body warmth will be stored by the suit's insulation, and

that will help somewhat. But you must not open the suit all night, not even to take off your helmet."

"And the oxygen?"

"You can take all the replacement cylinders from the ship and keep them at the satellite. The time you save by not having to go back and forth to the ship for fresh oxygen will give you about an hour's extra margin. You should be able to make it."

Tom nodded. "And of course I'm expected to work on the satellite right through the night."

"It will help you keep your mind off the cold. If we see that you're not going to make it—either because of the cold or the oxygen—we'll warn you and you can return to the settlement."

"Suppose I have enough oxygen to just finish the satellite, but if I do, I won't have enough to fly home. Will you warn me then?"

"Don't be dramatic."

"Go to hell."

"Dr. Arnoldsson said he could put you under," Jason continued unemotionally, "but he thinks you might freeze once your conscious mind went asleep."

"You've figured out all the details," Tom muttered. "All I have to do is put your damned satellite together without freezing to death and then fly 22,500 miles back home before my air runs out. Simple."

He glanced at the sun, still glaring bright even through his tinted visor. It was nearly on the edge of the Earth-disk.

"All right," Tom said, "I'm going into the ship now for some pills; it's nearly sunset."

Cold. Dark and so cold that numbers lost their meaning. Paralyzing cold, seeping in through the suit while you worked, crawling up your limbs until you could hardly move. The whole universe hung up in the sky and looked

down on the small cold figure of a man struggling blindly with machinery he could not understand.

Dark. Dark and cold.

Ruth stayed on the radio as long as Jason would allow her, talking to Tom, keeping the link with life and warmth. But finally Jason took over, and the radio went silent.

So don't talk, Tom growled silently, I can keep warm just by hating you, Jason.

He worked through the frigid night, struggling ant-like with huge pieces of equipment. Slowly he assembled the big parabolic mirror, the sighting mechanism and the atomic convertor. With dreamy motions he started connecting the intricate wiring systems.

And all the while he raged at himself: *Why? Why did it have to be this way? Why me? Why did I agree to do this? I knew I'd never live through it; why did I do it?*

He retraced the days of his life: the preparations for the flight, the arguments with Jason over exploring the cities, his trek from Chicago to the settlement, the aimless years after the radiation death of his two boys and Marjorie, his wife.

Marjorie and the boys, lying sick month after month, dying one after the other in a cancerous agony while he stood by helplessly in the ruins of what had been their home.

No! His mind warned him. Don't think of that. Not that. Think of Jason, Jason who prevents you from doing the one thing you want, who is taking your life from you; Jason, the peerless leader; Jason, who's afraid of the cities. Why? Why is he afraid of the cities? That's the hub of everything down there. Why does Jason fear the cities?

It wasn't until he finished connecting the satellite's last unit—the sighting mechanism—that Tom realized the answer.

One answer. And everything fell into place.

Everything ... except what Tom Morris was going to do about it.

Tom squinted through the twin telescopes of the sighting mechanism again, then pushed away and floated free, staring at the Earth bathed in pale moonlight.

What do I do now? For an instant he was close to panic, but he forced it down. *Think*, he said to himself. *You're supposed to be a Homo Sapiens ... use that brain. Think!*

The long night ended. The sun swung around from behind the bulk of Earth. Tom looked at it as he felt its warmth penetrating the insulated suit, and he knew it was the last time he would see the sun. He felt no more anger—even his hatred of Jason was drained out of him now. In its place was a sense of—finality.

He spoke into his helmet mike. "Jason."

"He is in conference with the astronomers." Dr. Arnoldsson's voice.

"Get him for me, please."

A few minutes of silence, broken only by the star-whisperings in his earphones.

Jason's voice was carefully modulated. "Tom, you made it."

"I made it. And the satellite's finished."

"It's finished? Good. Now, what we have to do...."

"Wait," Tom interrupted. "It's finished but it's useless."

"What?"

Tom twisted around to look at the completed satellite, its oddly-angled framework and bulbous machinery glinting fiercely in the newly-risen sun. "After I finished it I looked through the sighting mechanism to make certain the satellite's transmitters were correctly aimed at the settlement. Nobody told me to, but nobody said not to, either, so I looked. It's a simple mechanism.... The transmitters are pointed smack in the middle of Hudson's Bay."

"You're sure?"

"Certainly."

"You can rotate the antennas...."

"I know. I tried it. I can turn them as far south as the Great Lakes."

A long pause.

"I was afraid of this," Jason's voice said evenly.

I'll bet you were, Tom answered to himself.

"You must have moved the satellite out of position while assembling its components."

"So my work here comes to nothing because the satellite's power beam can't reach the settlement's receivers."

"Not ... not unless you use the ship ... to tow the satellite into the proper orbital position," Jason stammered.

You actually went through with it, Tom thought. Aloud, he said, "But if I use the ship's engine to tow the satellite, I won't have enough fuel left to get back to Earth, will I?" *Not to mention oxygen.*

A longer pause. "No."

"I have two questions, Jason. I think I know the answers to them both but I'll ask you anyway. One. You knew this would happen, didn't you?"

"What do you mean?"

"You've calculated this insane business down to the last drop of sweat," Tom growled. "You knew that I'd knock the satellite out of position while I was working on it, and the only way to get it back in the right orbit would be for me to tow it back and strand myself up here. This is a suicide mission, isn't it, Jason?"

"That's not true...."

"Don't bother defending yourself. I don't hate you anymore, Jason, I understand you, dammit. You made our deal as much to get rid of me as to get your precious satellite put together."

"No one can force you to tow the satellite...."

"Sure, I can leave it where it is and come back home. If I can fly this ship, which I doubt. And what would I come back to? I left a world without power. I'd return to a world without hope. And some dark night one of your disappointed young goons would catch up with me ... and no one would blame him, would they?"

Jason's voice was brittle. "You'll tow it into position?"

"After you answer my second question," Tom countered. "Why are you afraid of the cities?"

"Afraid? I'm not afraid."

"Yes you are. Oh, you could use the hope of exploring the cities to lure me up here on this suicide-job, but you knew I'd never be back to claim my half of the bargain. You're afraid of the cities, and I think I know why. You're afraid of the unknown quantity they represent, distrustful of your own leadership when new problems arise...."

"We've worked for more than ten years to make this settlement what it is," Jason fumed. "We fought and died to keep those marauding lunatics from wrecking us. We are mankind's last hope! We can't afford to let others in ... they're not scientists, they wouldn't understand, they'd ruin everything."

"Mankind's last hope, terrified of men." Tom was suddenly tired, weary of the whole struggle. But there was something he had to tell them.

"Listen Jason," he said. "The walls you've built around the settlement weren't meant to keep you from going outside. You're not a self-sufficient little community ... you're cut off from mankind's memory, from his dreams, from his ambitions. You can't even start to rebuild a civilization—and if you do try, don't you think the people outside will learn about it? Don't you think they've got a right to share in whatever progress the settlement makes? And if you don't let them, don't you realize that they'll destroy the settlement?"

Silence.

"I'm a historian," Tom continued, "and I know that a civilization can't exist in a vacuum. If outsiders don't conquer it, it'll rot from within. It's happened

to Babylonia, Greece, Rome, China, even. Over and again. The Soviets built an Iron Curtain around themselves, and wiped themselves out because of it.

"Don't you see, Jason? There are only two types of animals on this planet: the gamblers and the extinct. It won't be easy to live with the outsiders, there'll be problems of every type. But the alternative is decay and destruction. *You've got to take the chance, if you don't you're dead.*"

A long silence. Finally Jason said, "You've only got about a half-hour's worth of oxygen left. Will you tow the satellite into the proper position?"

Tom stared at the planet unseeingly. "Yes," he mumbled.

"I'll have to check some calculations with the astronomers," Jason's voice buzzed flatly in his earphones.

A background murmur, scarcely audible over the crackling static.

Then Ruth's voice broke through, "Tom, Tom, you can't do this! You won't be able to get back!"

"I know," he said, as he started pulling his way along the lifeline back to the ship.

"*No!* Come back, Tom, please. Come back. Forget the satellite. Come back and explore the cities. I'll go with you. Please. Don't die, Tom, please don't die...."

"Ruth, Ruth, you're too young to cry over me. I'll be all right, don't worry."

"No, it isn't fair."

"It never is," Tom said. "Listen, Ruth. I've been dead a long time. Since the bombs fell, I guess. My world died then and I died with it. When I came to the settlement, when I agreed to make this flight, I think we all knew I'd never return, even if we wouldn't admit it to ourselves. But I'm just one man, Ruth, one small part of the story. The story goes on, with or without me. There's tomorrow ... your tomorrow. I've got no place in it, but it belongs to you. So don't waste your time crying over a man who died eighteen years ago."

He snapped off his suit radio and went the rest of the way to the ship in silence. After locking the hatch and pumping air back into the cabin, he took off his helmet.

Good clean canned air, Tom said to himself. Too bad it won't last longer.

He sat down and flicked a switch on the radio console. "All right, do you have those calculations ready?"

"In a few moments...." Arnoldsson's voice.

Ten minutes later Tom re-emerged from the ship and made his ghost-like way back to the satellite's sighting mechanism. He checked the artificial moon's position, then went back to the ship.

"On course," he said to the radio. "The transmitters are pointing a little northwest of Philadelphia."

"Good," Arnoldsson's voice answered. "Now, your next blast should be three seconds' duration in the same direction...."

"No," Tom said, "I've gone as far as I'm going to."

"What?"

"I'm not moving the satellite any farther."

"But you still have not enough fuel to return to Earth. Why are you stopping here?"

"I'm not coming back," Tom answered. "But I'm not going to beam the satellite's power to the settlement, either."

"What are you trying to pull?" Jason's voice. Furious. Panicky.

"It's simple, Jason. If you want the satellite's power, you can dismantle the settlement and carry it to Pennsylvania. The transmitters are aimed at some good farming country, and within miles of a city that's still half-intact."

"You're insane!"

"Not at all. We're keeping our deal, Jason. I'm giving you the satellite's power, and you're going to allow exploration of the cities. You won't be able to prevent your people from rummaging through the cities now; and you won't be able to keep the outsiders from joining you, not once you get out from behind your own fences."

"You can't do this! You...."

Tom snapped off the radio. He looked at it for a second or two, then smashed a heavy-booted foot against the console. Glass and metal crashed satisfactorily.

Okay, Tom thought, it's done. Maybe Jason's right and I'm crazy, but we'll never know now. In a year or so they'll be set up outside Philadelphia, and a lot better for it, I'm forcing them to take the long way back, but it's a better way. The only way, maybe.

He leaned back in the seat and stared out the observation port at the completed satellite. Already it was taking in solar energy and beaming it Earthward.

In ten years they'll send another ship up here to check the gadget and make sure everything's okay. Maybe they'll be able to do it in five years. Makes no difference. I'll still be here.

THE END

*** END OF THE PROJECT GUTENBERG EBOOK A LONG WAY
BACK ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.